3) Handling Missing and Inappropriate Data in a Dataset

Program:

```
import numpy as np
import pandas as pd
df = pd.DataFrame({
    'CustomerID': [301, 302, 303, 304, 305, 306, 307, 308, 309, 309, 310],
    'Age_Group': ['18-23', '24-29', '30-35', '18-23', '36+', '36+',
                  '30-35', '18-23', '24-29', '24-29', '30-35'],
    'Rating(1-5)': [4, 2, 5, -2, 3, 7, 1, 4, 5, 5, 3],
    'Hotel': ['Taj', 'ITC', 'Marriott', 'ITC', 'Taj', 'Taj',
              'Marriott', 'ITC', 'Taj', 'Taj', 'Marriott'],
    'FoodPreference': ['Veg', 'Non-Veg', 'veg', 'Vegetarian', 'Veg',
                       'Non-Veg', 'Vegetarian', 'Veg', 'Non-Veg', 'Non-Veg', 'veg'],
    'Bill': [2200, 1800, 3200, -500, 1500, 2800, -900, 3500, 2400, 2400, 1900],
    'NoOfPax': [2, 4, 3, 5, 2, 22, -3, 6, 4, 4, 3],
    'EstimatedSalary': [52000, 68000, 75000, 85000, 62000, 71000,
                        -20000, 250000, 58000, 58000, 90000],
    'Age_Group.1': ['18-23', '24-29', '30-35', '18-23', '36+', '36+',
                    '30-35', '18-23', '24-29', '24-29', '30-35']
})
print("ORIGINAL DATASET")
print(df)
print("\nDUPLICATE ROWS")
print(df.duplicated())
df.drop_duplicates(inplace=True)
df.reset_index(drop=True, inplace=True)
df.drop(['Age_Group.1'], axis=1, inplace=True)
df.loc[
```

```
    (df['Rating(1-5)'] < 1) |
    (df['Rating(1-5)'] > 5),
    'Rating(1-5)'
] = np.nan
df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan
df.loc[
    (df['NoOfPax'] < 1) |
    (df['NoOfPax'] > 20),
    'NoOfPax'
] = np.nan
print("\nUNIQUE AGE GROUPS")
print(df['Age_Group'].unique())
print("\nUNIQUE HOTELS")
print(df['Hotel'].unique())
print("\nUNIQUE FOOD PREFERENCES")
print(df['FoodPreference'].unique())
df['FoodPreference'] = df['FoodPreference'].replace(
    ['veg', 'Vegetarian', 'Veg'],
    'Veg'
)
df['FoodPreference'] = df['FoodPreference'].replace(
    ['non-Veg', 'non-veg'],
    'Non-Veg'
)
df['EstimatedSalary'] = df['EstimatedSalary'].fillna(
    round(df['EstimatedSalary'].mean())
)
df['NoOfPax'] = df['NoOfPax'].fillna(
```

```
    round(df['NoOfPax'].median())
)

df['Rating(1-5)'] = df['Rating(1-5)'].fillna(
    round(df['Rating(1-5)'].median())
)
df['Bill'] = df['Bill'].fillna(
    round(df['Bill'].mean())
)
print("\nFINAL CLEANED DATASET")
print(df)
```

Output:

```
ORIGINAL DATASET
    CustomerID Age_Group  Rating(1-5)     Hotel FoodPreference  Bill  NoOfPax  \
0          301     18-23            4       Taj            Veg  2200        2
1          302     24-29            2       ITC        Non-Veg  1800        4
2          303     30-35            5  Marriott            veg  3200        3
3          304     18-23           -2       ITC     Vegetarian  -500        5
4          305       36+            3       Taj            Veg  1500        2
5          306       36+            7       Taj        Non-Veg  2800       22
6          307     30-35            1  Marriott     Vegetarian  -900       -3
7          308     18-23            4       ITC            Veg  3500        6
8          309     24-29            5       Taj        Non-Veg  2400        4
9          309     24-29            5       Taj        Non-Veg  2400        4
10         310     30-35            3  Marriott            veg  1900        3
```

```
    EstimatedSalary Age_Group.1
0             52000       18-23
1             68000       24-29
2             75000       30-35
3             85000       18-23
4             62000         36+
5             71000         36+
6            -20000       30-35
7            250000       18-23
8             58000       24-29
9             58000       24-29
10            90000       30-35

DUPLICATE ROWS
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
```

```
UNIQUE AGE GROUPS
<StringArray>
['18-23', '24-29', '30-35', '36+']
Length: 4, dtype: str

UNIQUE HOTELS
<StringArray>
['Taj', 'ITC', 'Marriott']
Length: 3, dtype: str

UNIQUE FOOD PREFERENCES
<StringArray>
['Veg', 'Non-Veg', 'veg', 'Vegetarian']
Length: 4, dtype: str
```

```
FINAL CLEANED DATASET
   CustomerID Age_Group  Rating(1-5)     Hotel FoodPreference    Bill  \
0         301     18-23          4.0       Taj            Veg  2200.0
1         302     24-29          2.0       ITC        Non-Veg  1800.0
2         303     30-35          5.0  Marriott            Veg  3200.0
3         304     18-23          4.0       ITC            Veg  2412.0
4         305       36+          3.0       Taj            Veg  1500.0
5         306       36+          4.0       Taj        Non-Veg  2800.0
6         307     30-35          1.0  Marriott            Veg  2412.0
7         308     18-23          4.0       ITC            Veg  3500.0
8         309     24-29          5.0       Taj        Non-Veg  2400.0
9         310     30-35          3.0  Marriott            Veg  1900.0

   NoOfPax  EstimatedSalary
0      2.0          52000.0
1      4.0          68000.0
2      3.0          75000.0
3      5.0          85000.0
4      2.0          62000.0
5      4.0          71000.0
6      4.0          90111.0
7      6.0         250000.0
8      4.0          58000.0
9      3.0          90000.0
```

Result:

Inappropriate data in the dataset is successfully studied.